Higher-Order Asymptotics of the Discrepancy Function under Fixed Model and Distributional Misspecification

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Purpose

The companion note `rmsea_asymptotics.tex` fixes the meaning of RMSEA by a Pitman drift: the misspecification is sent to zero at the $\sqrt{n}$ rate so the test statistic converges to a (noncentral) χ^2 and the noise inflation is the first-order Satorra–Bentler trace $\mathrm{tr}(U\Gamma)$. This note drops the drift. It expands the fitted discrepancy $\hat{F}$ to order n^{-1} under *fixed* misspecification in both the covariance structure (σ_0 off the model) and the distribution ($\Gamma \neq$ normal theory), for a general minimum-discrepancy estimator. The point is to see what the local sequence silently discards and to ask whether GLS-RMSEA differs from normal-theory RMSEA at all once the drift is removed.

The single result is the bias expansion

$$\mathbb{E}[\hat{F}] \;=\; F_0 \;+\; \frac{1}{n}\Big[\;\underbrace{\mathrm{tr}(\tilde{U}\Gamma)}_{\text{generalized } df}\;+\;\underbrace{\beta_W}_{\text{weight influence, } \propto e}\;\Big] \;+\; o(n^{-1}), \qquad (1)$$

with two objects that the drift kills. The trace carries the *curvature-corrected* residual-weight matrix $\tilde{U} = V - V\dot{\sigma}B^{-1}\dot{\sigma}'V$, built from the population *Hessian* $B = A - K$ of the discrepancy rather than its Gauss–Newton part $A = \dot{\sigma}'V\dot{\sigma}$; the two agree only when the residual–curvature coupling K vanishes (correct specification, or a model linear in θ). The remainder β_W is the influence of an estimated weight; it is identically zero for a fixed weight (ULS) and vanishes as the residual $e \to 0$ for any estimator. Three corollaries read off (1): a curvature-corrected, robust bias correction for RMSEA

(Corollary 1); an honest Wald confidence interval that needs no noncentral-χ^2 inversion (Corollary 2); and the ML/GLS/ULS comparison (Section 6).

Two structural remarks orient the rest. First, $\tilde{U}$ uses the Hessian B where the first-order projector U uses A; this is the fit-index sibling of the observed-Hessian bread that replaces Fisher information in the misspecification-robust standard errors of `misspec_observed_bread.tex`, and β_W is the fit-index sibling of the data-dependent-weight meat term there. Both corrections are zero at H_0 and leading-order off it; the order-promotion is the same theorem twice. Second, in the language of `aic_tic_rmsea_bias.tex`, $\mathrm{tr}(\tilde{U}\Gamma)$ is to $\mathrm{tr}(U\Gamma)$ as TIC's $\mathrm{tr}(I^{-1}J)$ is to AIC's k, and then one order deeper: the curvature term is a correction past even the first-order robust (TIC-like) trace.

1 Setup

Stack the $p^* = p(p+1)/2$ distinct sample covariances as $s = \mathrm{vech}\,\hat{\Sigma}$ (append means for a mean structure; the algebra is unchanged), with population limit $s \xrightarrow{p} \sigma_0$ and

$$z := \sqrt{n}\,(s - \sigma_0) \xrightarrow{d} N(0, \Gamma), \tag{2}$$

Γ the asymptotic covariance of the moments, normal-theory or not. The model is a smooth map $\theta \mapsto \sigma(\theta)$, $\theta \in \mathbb{R}^t$, $df = p^* - t$. A minimum-discrepancy estimator minimizes a quadratic form in the residual $r(\theta) = s - \sigma(\theta)$,

$$\hat{F} = \min_{\theta}\ r(\theta)'\hat{V}\,r(\theta), \qquad \hat{V} \xrightarrow{p} V, \tag{3}$$

with weight $\hat{V}$ either fixed (ULS) or a smooth function of the data (GLS). Maximum likelihood is not exactly of this form; it is treated as the reweighted limit in Section 6.

Pseudo-true value and population residual. Let

$$\theta_* = \arg\min_{\theta}\,(\sigma_0 - \sigma(\theta))'V(\sigma_0 - \sigma(\theta)), \qquad e := \sigma_0 - \sigma(\theta_*), \qquad F_0 = e'Ve. \tag{4}$$

Write $\dot{\sigma} = \partial\sigma/\partial\theta'$ at θ_* ($p^* \times t$, full column rank) and the model Hessian tensor $\ddot{\sigma}$, $(\ddot{\sigma})_{a,ij} = \partial^2\sigma_a/\partial\theta_i\partial\theta_j$. The first-order condition for (4) is the V-orthogonality of the residual to the tangent space,

$$\dot{\sigma}'Ve = 0. \tag{5}$$

Under correct specification $e = 0$; under fixed misspecification $e = O(1)$ and (5) is all that survives of it.

Curvature objects. Define the Gauss–Newton matrix, the residual–curvature coupling, and the population Hessian of the discrepancy,

$$A := \dot\sigma' V \dot\sigma, \qquad K_{ij} := (Ve)'(\ddot\sigma)_{:,ij} = \sum_a (Ve)_a (\ddot\sigma)_{a,ij}, \qquad B := A - K = \tfrac{1}{2}\, \partial^2_{\theta\theta}\big(e'Ve\big). \tag{6}$$

K is the only object in (6) that requires both a nonzero residual (Ve) and a curved model ($\ddot\sigma \neq 0$); it is the misspecification signature. With B in hand define the standard residual projector and its curvature-corrected analogue

$$U := V - V\dot\sigma A^{-1}\dot\sigma' V, \qquad \tilde U := V - V\dot\sigma B^{-1}\dot\sigma' V = V - VPV, \quad P := \dot\sigma B^{-1}\dot\sigma'. \tag{7}$$

U is a genuine residual projector ($U\dot\sigma = 0$, idempotent in the V-metric); $\tilde U$ is a perturbation of it that no longer annihilates the tangent ($\tilde U\dot\sigma = -V\dot\sigma B^{-1}K$), reducing to U as $K \to 0$. U is the object of every limit law in the drift regime (`rmsea_asymptotics.tex`); $\tilde U$ is its fixed-misspecification replacement.

Weight fluctuation. For an estimated weight expand

$$\hat V = V + n^{-1/2}\dot V + \tfrac{1}{2} n^{-1}\ddot V + o_p(n^{-1}), \tag{8}$$

where $\dot V = O_p(1)$ is linear in z (the derivative of the weight map along the moment fluctuation) and $\ddot V = O_p(1)$ is the second-order term. Write the residual-projected weight fluctuation

$$w := \dot V\, e, \tag{9}$$

an $O_p(1)$ vector that is identically zero for a fixed weight and is $O(\|e\|)$ otherwise. For GLS, $\dot V$ is driven by the raw moment fluctuation z; for ML, by the projected fluctuation $\dot\sigma\,(\hat\theta - \theta_*)$ (Section 6).

2 The stochastic expansion

Theorem 1 (Second-order expansion of $\hat F$)**.** *Assume (2), identification ($\dot\sigma$ full rank, B nonsingular), a weight that does not depend on θ (ULS, GLS, or any fixed/data weight), and the usual smoothness. Then*

$$\hat F = F_0 + n^{-1/2}\left(2e'Vz + e'\dot V e\right) + n^{-1}\,\Phi + o_p(n^{-1}), \tag{10}$$

with

$$\Phi = z'\tilde{U}z \;+\; 2\,w'(I-PV)z \;-\; w'Pw \;+\; \tfrac{1}{2}\,e'\ddot{V}e. \tag{11}$$

Proof. Write $u = \hat{\theta} - \theta_* = O_p(n^{-1/2})$ and split it as $u = n^{-1/2}\xi + O_p(n^{-1})$. Expand the residual at the estimator,

$$r(\hat{\theta}) = e + \rho_1 + \rho_2 + o_p(n^{-1}), \quad \rho_1 = n^{-1/2}z - \dot{\sigma}\,u, \quad \rho_2 = -\tfrac{1}{2}\,(\ddot{\sigma})[u,u],$$

where $(\ddot{\sigma})[u,u]$ has a-component $u'(\ddot{\sigma})_{:,a}u$. The estimating equation $\dot{\sigma}(\hat{\theta})'\hat{V}r(\hat{\theta}) = 0$, expanded to $O_p(n^{-1/2})$ with $\dot{\sigma}(\hat{\theta}) = \dot{\sigma} + (\ddot{\sigma})[u,\cdot]$ and (8), gives after using (5)

$$-A\,u + (\ddot{\sigma})[u,\cdot]'Ve + n^{-1/2}\dot{\sigma}'Vz + n^{-1/2}\dot{\sigma}'\dot{V}e = o_p(n^{-1/2}).$$

The second term is Ku by (6); collecting,

$$(A-K)\,u = n^{-1/2}\dot{\sigma}'(Vz+\dot{V}e)+o_p(n^{-1/2}) \implies \xi = B^{-1}\dot{\sigma}'(Vz+w), \quad u = n^{-1/2}\xi. \tag{12}$$

The weight fluctuation enters ξ only through $w = \dot{V}e$, hence only under misspecification. Now substitute into $\hat{F} = (e+\rho)'\hat{V}(e+\rho)$, $\rho = \rho_1 + \rho_2$, and use (5) twice. The cross term with ρ_1 collapses to the linear part, $2e'V\rho_1 = 2n^{-1/2}e'Vz$ (the $\dot{\sigma}u_1$ piece dies by (5)); the cross term with ρ_2 kills the second-order parameter increment and leaves only curvature, $2e'V\rho_2 = -u'Ku$; and $e'\hat{V}e$ contributes the weight terms $n^{-1/2}e'\dot{V}e + \frac{1}{2}n^{-1}e'\ddot{V}e$. Together with $\rho_1'V\rho_1 = n^{-1}(z'Vz - 2\xi'\dot{\sigma}'Vz + \xi'A\xi)$ and the $\dot{V}$–ρ_1 cross term $2n^{-1}\,w'\big(z - \dot{\sigma}\xi\big)$, the $O(n^{-1})$ part is

$$\Phi = z'Vz - 2\xi'\dot{\sigma}'Vz + \xi'B\xi + 2w'(z - \dot{\sigma}\xi) + \tfrac{1}{2}e'\ddot{V}e,$$

where $\xi'A\xi - u'Ku/n^{-1}$ combined to $\xi'B\xi$. Insert $\xi = B^{-1}\dot{\sigma}'(Vz+w)$. With $P = \dot{\sigma}B^{-1}\dot{\sigma}'$ one gets $\xi'B\xi = (Vz+w)'P(Vz+w)$, $\xi'\dot{\sigma}'Vz = (Vz+w)'PVz$ and $\dot{\sigma}\xi = P(Vz+w)$. Expanding and cancelling,

$$\Phi = z'(V - VPV)z + 2w'(I-PV)z - w'Pw + \tfrac{1}{2}e'\ddot{V}e,$$

which is (11) with $\tilde{U} = V - VPV$. □

Remark (The cancellation is the content). Everything hinges on the population FOC (5). It removes the tangent-direction sampling noise from the linear term (leaving $2e'Vz = 2e'Uz$, since $e'V = e'U$ by (5)) and removes the second-order parameter increment u_2 from the bias, so the bias needs ξ only to first order. What remains is genuinely a residual-space object, projected by $\tilde{U}$.

3 Bias and the curvature-corrected df

Corollary 1 (Generalized degrees of freedom)**.** *Under the conditions of Theorem 1,*

$$\mathbb{E}[\hat{F}] = F_0 + \frac{1}{n}\big[\operatorname{tr}(\tilde{U}\Gamma) + \beta_W\big] + o(n^{-1}), \qquad \beta_W = 2\mathbb{E}\big[w'(I - PV)z\big] - \mathbb{E}\big[w'Pw\big] + \tfrac{1}{2}e'\mathbb{E}[\ddot{V}]\,e. \tag{13}$$

For a fixed weight $\beta_W \equiv 0$; for any estimator $\beta_W = O(\|e\|)$, so it vanishes at H_0. Under correct specification $e = 0$, $K = 0$, $\tilde{U} = U$ and (13) *reduces to $\mathbb{E}[\hat{F}] = \operatorname{tr}(U\Gamma)/n + o(n^{-1})$, the Satorra–Bentler mean $c\,df = \operatorname{tr}(U\Gamma)$ of* `rmsea_asymptotics.tex`*, which is df when $V = \Gamma^{-1}$.*

Proof. Take expectations in (10). The $n^{-1/2}$ term has mean zero ($\mathbb{E}z = 0$, $\mathbb{E}\dot{V} = 0$). For the n^{-1} term, $\mathbb{E}[z'\tilde{U}z] = \operatorname{tr}(\tilde{U}\Gamma)$ by the quadratic-form mean and the remaining terms define β_W. □

The estimand correction therefore generalizes the chain

$$df \;\longrightarrow\; \operatorname{tr}(U\Gamma) \;\longrightarrow\; \operatorname{tr}(\tilde{U}\Gamma) + \beta_W,$$

in increasing order of fidelity: df assumes a correctly weighted, correctly distributed null; $\operatorname{tr}(U\Gamma)$ allows distributional misspecification but freezes the model at its tangent; $\operatorname{tr}(\tilde{U}\Gamma) + \beta_W$ allows fixed structural misspecification, correcting the projector for curvature and adding the weight influence. A curvature-and-weight corrected RMSEA is

$$\hat{\varepsilon}_{\text{ho}} = \sqrt{\max\Big(\frac{\hat{F}}{df} - \frac{\operatorname{tr}(\hat{\tilde{U}}\hat{\Gamma}) + \hat{\beta}_W}{n\,df},\, 0\Big)}. \tag{14}$$

Remark (Curvature is real for SEM). $K \neq 0$ requires $\ddot{\sigma} \neq 0$. Factor and path models are nonlinear in θ (loadings multiply, paths compose), so $\ddot{\sigma} \neq 0$ generically and the curvature term is present whenever the model is wrong. It vanishes for models linear in θ (e.g. a free covariance matrix with linear equality constraints), where $\tilde{U} = U$ exactly even under misspecification.

4 The honest interval

Corollary 2 (CLT and Wald interval, no noncentral inversion)**.** *If $F_0 > 0$, then from* (10)

$$\sqrt{n}\,(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2), \qquad \omega^2 = \operatorname{Var}\big(2e'Vz + e'\dot{V}e\big) = (2Ve + v_e)'\Gamma(2Ve + v_e), \tag{15}$$

where v_e is the vector representing the linear functional $z \mapsto e'\dot{V}e = v_e'z$; for a fixed weight $v_e = 0$ and $\omega^2 = 4e'V\Gamma Ve = 4e'U\Gamma Ue$. By the delta method, with $\varepsilon = \sqrt{F_0/df}$,

$$\sqrt{n}\,(\widehat{\varepsilon} - \varepsilon) \xrightarrow{d} N\Big(0,\ \frac{\omega^2}{4\,df\,F_0}\Big), \tag{16}$$

giving a Wald interval for ε directly on the discrepancy scale.

Equation (16) is the fixed-misspecification interval: under a genuinely wrong model the statistic is asymptotically normal, not noncentral χ^2, and the interval is a plain sandwich, not a CDF inversion. It complements rather than replaces the noncentral interval of `rmsea_asymptotics.tex`: that one is built for the near-null regime ($F_0 \downarrow 0$) where (16) degenerates (ω^2/F_0 blows up), while (16) is built for clearly-misspecified models where the noncentral inversion is least trustworthy. The crossover is exactly where neither device is comfortable, which is the honest statement of where RMSEA interval coverage is hard.

5 What the drift discards

Put back the Pitman sequence $e_n = n^{-1/2}\delta$ with δ a fixed off-model direction, the regime of `rmsea_asymptotics.tex`. Then $Ve_n = O(n^{-1/2})$ so $K = O(n^{-1/2}) \to 0$, hence $B \to A$ and $\tilde{U} \to U$; the weight influence $w = \dot{V}e_n = O_p(n^{-1/2}) \to 0$, so $\beta_W \to 0$; and the normal CLT (15) degenerates while $n\hat{F}$ acquires the (noncentral) χ^2 limit. The drift annihilates precisely the two corrections this note is about.

	fixed misspecification ($e = O(1)$)	Pitman drift ($e = n^{-1/2}\delta$)
limit law of $\hat{F}$	normal, $\sqrt{n}(\hat{F} - F_0) \xrightarrow{d} N(0, \omega^2)$	$n\hat{F} \xrightarrow{d}$ (noncentral) χ^2 mixture
noise/bias mean	$\mathrm{tr}(\tilde{U}\Gamma)$, curvature-corrected	$\mathrm{tr}(U\Gamma) = c\,df$, no curvature
weight influence β_W	$O(\lVert e\rVert)$, present	$\to 0$
ML vs GLS	differ (two-metric, pseudo-true)	coincide to leading order
interval	Wald via ω^2 (16)	noncentral-χ^2 inversion

The local-asymptotics motivation is not wrong; it is a deliberate choice to work in the only regime where the *df* subtraction is exact and the curvature and weight terms are negligible. Outside it, those terms are the leading description of a wrong model's fit.

6 ML, GLS, ULS

The estimators differ in where the weight is anchored, and once the drift is gone this separates their RMSEAs. The unifying tool is that $\tilde{U}$ is, for *any* smooth discrepancy, half the Hessian of the profiled criterion.

Lemma 1 (Profiled-Hessian form)**.** *For a smooth discrepancy $F(s,\theta)$ with population minimizer θ_*, Corollary 1 holds with*

$$\tilde{U} \;=\; \tfrac{1}{2}\nabla_s^2\Big[\min_\theta F(s,\theta)\Big]\Big|_{s=\sigma_0} \;=\; V_{ss} - \tfrac{1}{4}\,C\,B^{-1}C', \tag{17}$$

where $V_{ss} = \frac{1}{2}\partial^2_{ss}F$, $C = \partial^2_{s\theta}F$ and $B = \frac{1}{2}\partial^2_{\theta\theta}F$ at (σ_0, θ_*). The fixed-weight quadratic (7) *is the case $V_{ss} = V$, $C = -2V\dot{\sigma}$ (then $\frac{1}{4}CB^{-1}C' = V\dot{\sigma}B^{-1}\dot{\sigma}'V$). A non-quadratic F enters only through these three second derivatives; there is no separate higher-derivative term at $O(1/n)$, and the data skewness enters the bias only at $O(n^{-3/2})$. For a data-dependent weight $V(s)$ the s-derivatives of the weight populate V_{ss} and C, reproducing β_W of Corollary 1 in trace form.*

ULS. Fixed weight, $\dot{V} = 0$, so $\beta_W \equiv 0$ and the bias is purely $\mathrm{tr}(\tilde{U}\Gamma)/n$. ULS is never efficient, so $\mathrm{tr}(U\Gamma) \neq df$ already at leading order regardless of normality; the curvature term is then a clean second correction with no weight bookkeeping. This is the cleanest place to exhibit $\tilde{U} \neq U$.

GLS. Weight $\hat{V} = V(s)$ at the sample moments, so $\dot{V}$ is the derivative of $V(\cdot)$ along the *raw* fluctuation z. The full β_W is present, leading-order in the residual through the cross term $2\mathbb{E}[w'(I-PV)z]$, and ω^2 in (15) acquires the v_e term.

ML. ML is not a fixed quadratic: its normal equation $\dot{\sigma}'W(\theta)\,(\sigma_0-\sigma(\theta)) = 0$ anchors the weight $W(\theta) = \frac{1}{2}D'(\Sigma(\theta)^{-1} \otimes \Sigma(\theta)^{-1})D$ at the *fitted model*. Two consequences. First, the pseudo-true value solves a model-metric projection, so under fixed misspecification $\theta_*^{\mathrm{ML}} \neq \theta_*^{\mathrm{GLS}}$ and $F_0^{\mathrm{ML}} \neq F_0^{\mathrm{GLS}}$: the two estimate *different population numbers*, already at zeroth order, measuring misfit in different metrics (Olsson, Foss, Troye and Howell [2]). Second, by Lemma 1 the noise weight is the moment-Hessian $V_{ss} = V_0 =$

$\frac{1}{2}D'(\Sigma_0^{-1} \otimes \Sigma_0^{-1})D$ at the *data*, while the cross-term $C = -2W_*\dot{\sigma}$ carries the *model* weight $W_* = \frac{1}{2}D'(\Sigma_*^{-1} \otimes \Sigma_*^{-1})D$, $\Sigma_* = \Sigma(\theta_*)$, so

$$\tilde{U}_{\mathrm{ML}} = V_0 - W_*\dot{\sigma}\, B_{\mathrm{ML}}^{-1}\dot{\sigma}' W_*. \tag{18}$$

This is the two-metric object: sampling noise weighted in the data metric, residuals projected in the model metric. The non-quadraticity of the log-det is absorbed entirely into $(V_0, W_*, B_{\mathrm{ML}})$; there is no separate third-derivative term (correcting an earlier draft). It all collapses under correct specification ($\Sigma_* = \Sigma_0$, $W_* = V_0$, $e = 0$), where ML, GLS and ULS coincide to the order that matters. The numerical check confirms (18) to finite-difference precision.

So the answer to "does GLS-RMSEA do anything that normal-theory RMSEA does not" is yes, and sharper than expected: under fixed misspecification the two already target different estimands (different metric), and on top of that carry different $O(1/n)$ biases through β_W and B. Under the drift, all of this disappears and the classical equivalence is restored. The daylight between estimators lives entirely in the fixed-misspecification, higher-order regime.

7 Numerical check

A one-factor model is fit to a two-factor population (six indicators, three per factor, factor correlation 0.55), a misspecification the one-factor model cannot absorb ($df = 9$; ULS $F_0 = 0.23$, ML $F_0 = 0.49$, the differing estimands of Section 6). Two checks.

Deterministic. By Lemma 1 the bias coefficient is half the Hessian of the profiled criterion, which finite differencing pins exactly, with no Monte Carlo. Comparing $\frac{1}{2}\nabla_s^2[\min_\theta F]$ at σ_0 to the closed forms:

	$\\|\frac{1}{2}H_{\mathrm{prof}} - \tilde{U}\\|/\\|\tilde{U}\\|$	$\mathrm{tr}(\tilde{U}\Gamma_{\mathrm{NT}})$ corrected	naive (Gauss–Newton A)
ULS	2×10^{-8}	2.26	3.38
ML	9×10^{-6}	1.24	9.07

ULS is a single-metric quadratic (positive control); ML is the two-metric $\tilde{U}_{\mathrm{ML}}$ of (18), confirmed to finite-difference precision. Replacing the Hessian B by Gauss–Newton A is wrong by a factor of several, for ML by 7× (and with the opposite conclusion about how much sampling noise a wrong model spends): discarding the curvature is not a small error.

Monte Carlo (ULS, $\beta_W = 0$). Confirms the Hessian-to-bias step, $N(\mathbb{E}[\hat{F}] - F_0) \to \mathrm{tr}(\tilde{U}\Gamma)$, under normal and t_6 data; the sample covariance is the unbiased $\div(N-1)$ form (the $\div N$ form carries its own $O(1/N)$ moment bias contaminating this order), Γ_{NT} closed form under normality and Monte-Carlo under t_6:

data	N	$N(\mathbb{E}[\hat{F}] - F_0)$ (MC se)	naive $\mathrm{tr}(U\Gamma)$	corrected $\mathrm{tr}(\tilde{U}\Gamma)$
normal	200	2.23 (0.05)	3.38	**2.26**
normal	800	2.21 (0.10)	3.38	**2.26**
t_6	200	4.22 (0.08)	7.00	**4.74**
t_6	800	4.78 (0.16)	7.00	**4.74**

By $N = 800$ the empirical bias is within one Monte-Carlo standard error of $\mathrm{tr}(\tilde{U}\Gamma)$ in both conditions; the naive trace is off by the full gap (over twenty standard errors at $N = 200$, normal). ML is validated deterministically rather than by Monte Carlo: its bias has a large higher-order tail and small-N improper-solution inflation that the exact Hessian check sidesteps. The curvature term is not a third-decimal refinement; it is a third to a half of the whole bias. Script: `higher_order_discrepancy_check.py`.

8 Connections and caveats

Remark (Hessian bread, weight meat: the SE sibling). $\tilde{U}$ uses the population Hessian $B = A - K$ where U uses the Gauss–Newton A; under correct specification $K = 0$ and they agree, the discrepancy-function analogue of the information equality. This is the same substitution that the misspecification-robust standard errors of `misspec_observed_bread.tex` make when they replace the Fisher bread by the observed Hessian, and β_W is the fit-index counterpart of their data-dependent-weight meat term. Both corrections are $O(\|e\|)$, hence zero at H_0 and leading-order off it: the order-promotion (Newey–McFadden null irrelevance, Hall–Inoue off-null promotion) is one phenomenon appearing in the variance there and in the bias here.

Remark (TIC analogy, one order deeper). `aic_tic_rmsea_bias.tex` reads *df* as the AIC penalty and notes the robust trace as its TIC analogue. Here $\mathrm{tr}(U\Gamma)$ is that TIC-like trace (first order), and $\mathrm{tr}(\tilde{U}\Gamma) + \beta_W$ is the next order. The user's instinct stands: the *df* correction is AIC-like (a fixed integer), the robust trace is TIC-like (a sandwich trace), and the practical complaint about TIC, that $\mathrm{tr}(\hat{I}^{-1}\hat{J})$ is noisy because $\hat{J}$ is a fourth-moment object, transfers verbatim: $\mathrm{tr}(\hat{\tilde{U}}\hat{\Gamma})$ inherits the sampling variance of $\hat{\Gamma}$. The higher-order expansion is what lets one judge the trade: when $\|e\|$ and the kurtosis are small the AIC-like *df* is nearly unbiased and lower variance; when they are not, the bias of the cheap correction is exactly (13) and may dominate. A structured (model-implied or shrunk) $\hat{\Gamma}$ is the natural middle estimator, the RMSEA analogue of a regularized TIC.

Remark (Caveats). The truncation $\max(\cdot, 0)$ in (14) re-introduces an $O(n^{-1/2})$ bias near $F_0 = 0$, where the fixed-misspecification expansion is least relevant and the near-null device should take over. Data skewness enters the bias only at $O(n^{-3/2})$ (Lemma 1); the t_6 shortfall at $N = 200$ in the table is that tail. The expansion assumes finite fourth moments for Γ and finite sixth moments for the n^{-1} bias to have finite expectation; under heavier tails (13) is a stochastic-order statement, not a moment statement.

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